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Paleomagnetic constraints on the tectonic relationship between the Alashan/Hexi Corridor terrane and the North China Block

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Abstract. We report paleomagnetic results for Early-Middle Ordovician limestone rocks from the eastern part of the Hexi Corridor, northwest China. The characteristic high temperature component B isolated from 74 out of 114 samples exhibited dual polarity and passed the fold and reversal tests at the 95% confidence level. The corresponding paleopole, at 326.5°E, 31.8°N with A95=9.5°, is in agreement with the coeval poles from the stable North China Block (NCB). It implies that the Alashan/Hexi Corridor (AHC) terrane was part of the NCB since the Early-Middle Ordovician, which is consistent with geological evidence. The Silurian and Devonian paleomagnetic results from the AHC terrane may therefore be used to discuss the middle Paleozoic tectonic displacement and paleogeographic reconstruction of the NCB.

Introduction

The relationship between the AHC terrane and the stable NCB, which is defined as the eastern part of the NCB beyond the Helanshan fold belt, is an unresolved problem for geologists and paleomagnetists who are interested in the Paleozoic paleogeography of the NCB. Geological evidence seems to favor that the AHC area had been the western extension of the NCB since the pre-Cambrian, and accreted onto the Ordos terrane by collision and amalgamation during the Late Cambrian (Wu et al., 1993; Meng et al., 1990, 1992). The fine consistency of the Late Permian paleomagnetic results from the Jiuquan Basin of AHC terrane and the Ordos Basin of the stable NCB indicates that the AHC terrane had been the western extension of the NCB since the Late Permian (Meng et al., 1990). Besides the AHC terrane, however, Silurian and Devonian sediments were undeveloped in the eastern part of the NCB due to crustal uplift. This situation leads to a confused tectonic relationship and geographic configuration between the AHC terrane and the stable NCB. With the goal of better understanding the relationship between the AHC terrane and the stable NCB during the Ordovician, we sampled the Early-Middle Ordovician limestones from the southwest Zhongning County (37.2°N, 105.5°E) of the eastern part of the Hexi Corridor, southwest Ningxia Autonomous Region of China.

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Geological outline and sampling

The AHC terrane, which is situated in the west of the Ordos Basin through the Helanshan fold belt, is at the juncture of two major Chinese blocks, North China and Tarim (Fig. 1). This terrane is bounded by the Central Asian Fold Belt on the north and separated from Qaidam terrane by the Qilianshan Fold Belt to the south. The western boundary of the AHC terrane is not clearly delineated but is generally thought to be in the vicinity of northeastern extensions of the Altyn fault zone beyond northern Qilianshan.

The Hexi Corridor area is located in the southern margin of the AHC terrane. Two giant faults, North Qilian fault and Longshushan fault, running NWW-SEE on both southern and northern sides, control this area. Early Paleozoic deposition includes the well developed Middle Cambrian and Lower-Middle Ordovician series, which unconformably or disconformably contact both the overlying and underlying strata (NBGMR, 1990). Unconformity between the Late Silurian to Middle Devonian and the Late Devonian to Early Carboniferous in the AHC terrane indicates that Caledonian folding occurred in this terrane.

The Lower-Middle Ordovician series is composed of the Tianjinshan and Miboshan Formations in the southwest of

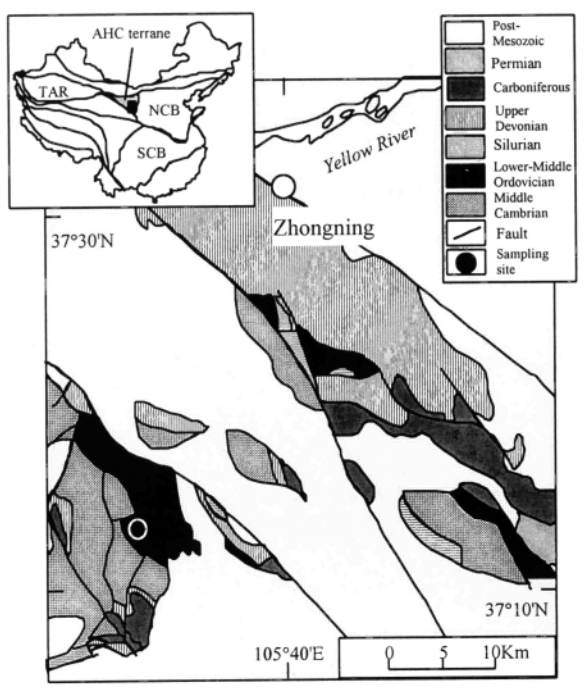


Figure 1. Sketch geological map of the Zhongning area of southwest Ningxia showing the Paleozoic geological stratigraphy and the paleomagnetic sampling site. AHC terrane, Alashan/Hexi Corridor terrane.

Table 1. Site Statistics of the Characteristic Remanent Magnetization (Component B)

Site ID	Strike/dip	n_1/n_2	N/R	D_g	I_g	D_s	I_s	κ	α_{95}	Slong	SLat
***Tianjinshan Formation** (105.5°E/37.2°N), limestone*											
YO48	121/43	8/9	8/0	326.2	-20.6	333.1	0.8	59.5	7.2	325.8	45.6
YO49	121/43	2/10	1/1	305.4	-32.1	326.9	-20.1	-	-	325.4	33.2
YO51	130/39	9/11	3/6	131.6	12.8	139.3	8.7	31.6	9.3	157.0	-33.8
YO53	117/36	2/4	2/0	350.7	-17.4	351.7	12.2	-	-	301.2	58.1
YO54	123/43	7/9	0/7	120.8	43.5	154.8	31.5	15.0	14.8	133.8	-30.8
YO55	131/43	9/11	0/9	92.6	29.3	125.8	46.7	67.2	7.2	151.8	-7.4
YO56	132/39	4/9	0/4	97.2	21.7	118.8	38.4	92.7	9.6	160.8	-7.7
YO57	129/39	11/13	0/11	116.4	48.9	158.5	42.5	28.5	8.7	127.1	-25.0
Miboshan Formation (105.5°E/37.2°N), limestone and oolitic limestone											
YO58	90/84	7/7	2/5	41.2	48.6	147.9	35.0	167.8	4.9	139.4	-25.9
YO60	88/86	7/8	7/0	246.1	-44.5	313.8	-18.3	29.7	11.3	338.2	26.4
YO63	118/57	3/5	0/3	142.9	21.8	149.9	-7.3	71.7	14.7	152.2	-46.6
YO64	139/79	2/9	0/2	125.5	-7.9	128.4	11.5	-	-	165.3	-25.5
YO65	147/64	3/9	3/0	314.5	-18.1	338.5	-18.7	135.5	10.6	313.2	39.0
Averaging by sites			C	D	I	κ	α_{95}	Long	Lat	K	A_{95}
Normal		5/13	G	313.8	-30.4	6.0	34.1	334.6	19.1	5.1	37.7
			S	332.9	-9.1	16.6	19.3	322.6	41.1	26.4	15.2
Reversed		8/13	G	112.4	30.3	6.6	23.3	168.7	-5.0	6.9	22.8
			S	140.6	26.7	12.7	16.2	148.5	-25.9	20.8	12.4
Mixed		13/13	G	300.6	-30.8	6.4	17.7	343.6	10.4	6.1	18.3
			S	325.7	-19.9	12.1	12.4	326.5	31.8	19.9	9.5

Site ID, site identification, Strike/dip: strike azimuth and dip of bed; n_1/n_2: number of samples, sites used in statistic analysis/total of samples, sites measured; N/R, normal/reversed; D_g/I_g (D_s/I_s), declination and inclination in situ (after tilt correction); C=coordinates, G=Geographic; S=Stratigraphic; Long/Lat, longitude and latitude of VGP.

Ningxia, while the early Early Ordovician Dananci Fm is absent there. The rocks of these two formations consist largely of limestones and oolitic limestones of dark or light gray color. Some critical fossils, e.g. cephalopods (*Coreanoceras, Kaipingoceras, Cyrtovaginoceras*), trilobites (*Pseudoasaphus, Megistaspia, Symphysurus,* and *Illaenus*), gastropods (*Maclurites, Ophileta, Tropidodiscus*), and conodonts (*Bergstroemognathus, Baltoniodus*), indicate that the Tianjinshan Fm is correlative with the Honghuayuanian and Dawanian stages (Arenig Stage, NBGMR, 1990; Yang et al., 1986). The Miboshan Fm is characterized by the abundance of graptolites (*Amplexograptus confertus Zone, Pterograptus elegans Zone*) indicative of Guniutanian age (Llanvirn Stage in Europe, NBGMR, 1990).

In the sampling area, the Tianjinshan and Miboshan formations contact each other with conformity whereas they contact the underlying and overlying strata both with unconformity or fault (Fig. 1). Thirteen sites consisting of about 10 limestone samples per site were collected from a single monoclinal section with a portable drill and oriented using a magnetic compass. The bedding attitudes are similar in strike but with quite different dips between the two formations (Table 1).

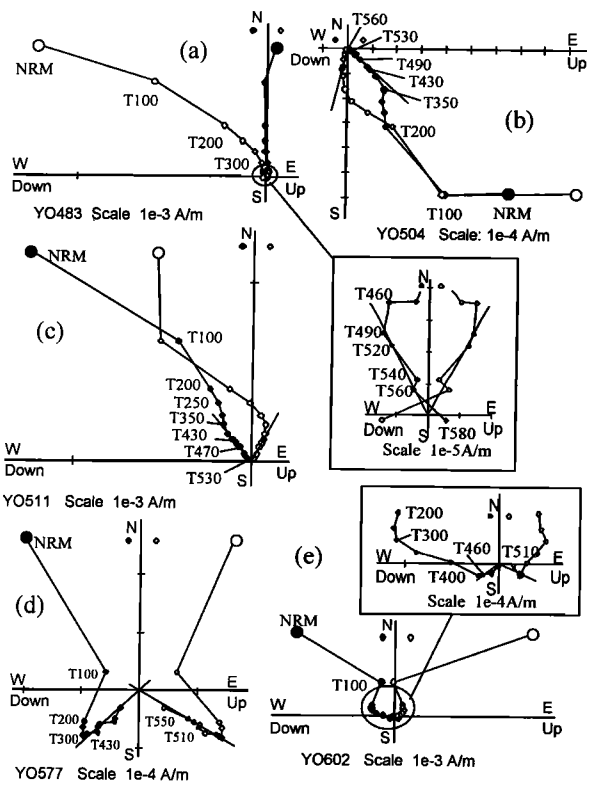


Figure 2. Orthogonal vector plots of representative samples. Directions are plotted in-situ, open and solid circles represent vector endpoints projected onto the vertical and horizontal planes respectively. T indicates thermal (°C) demagnetization.

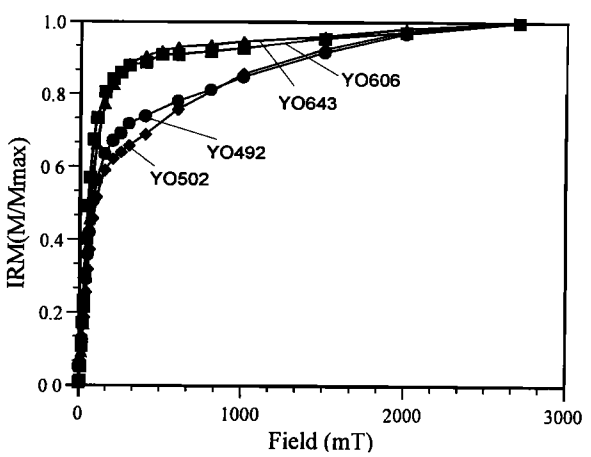


Figure 3. Plots of some representative IRM acquisition curves for the Early-Middle Ordovician limestones.

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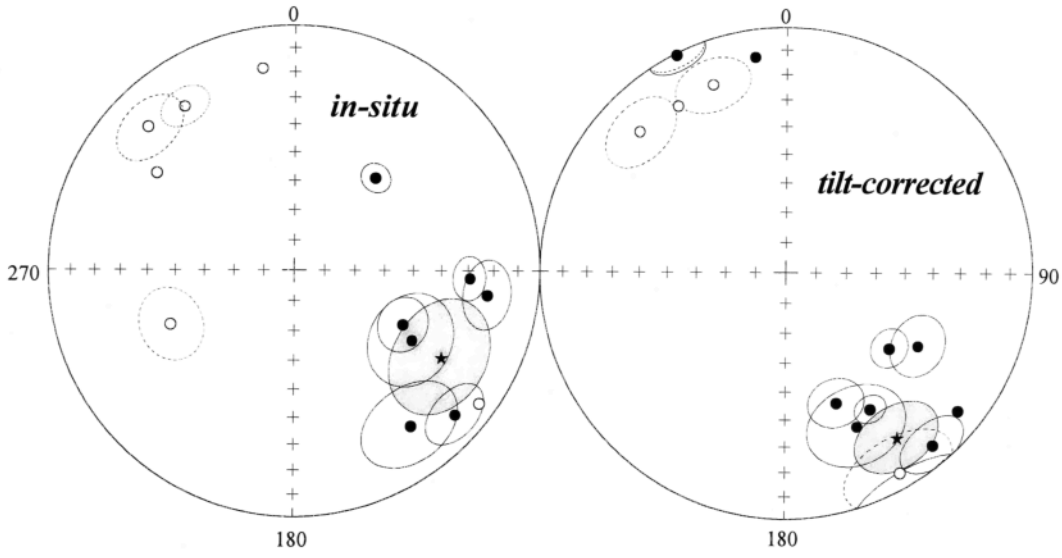


Figure 4. Equal-area projection of the site-mean characteristic remanent magnetization (component B) directions before and after tilt correction, showing the normal and reversed directions and the grouping of characteristic directions. Solid/open symbols represent directions plotted onto the lower/upper hemisphere with 95% confidence limits. Stars, mean directions.

Results

Except for a few samples which were subjected to hybrid thermal and AF demagnetization, all the samples were progressively thermally demagnetized. Remanent magnetization was measured with a 2G Enterprises cryogenic magnetometer at Kobe University, Japan. Standard analytical techniques (Zijderveld, 1967; Kirschvink, 1980; Fisher, 1953) were employed to analyze the demagnetization results.

Most of the samples yielded two components after the removal of a recent viscous overprint at 100°C or 200°C (Fig. 2). Predominant northerly steep downward and a few reversed directions (3 samples) (in-situ), designated as component A hereafter, were separated in temperature range of 200 to 350°C. A characteristic high temperature component, designated as component B, was revealed between 400°C and 580°C or 590°C (Fig. 2). This component was isolated after 30mT using a hybrid demagnetization method. IRM acquisition experiments (Fig. 3) and the demagnetization behavior confirm that this component resides in magnetite. In some cases, however, component A almost obliterates component B, and demagnetization revealed a distinct great-circle trend toward the characteristic direction but failed to isolate a stable end-point direction. Site statistics of component B are given in Table 1.

Component A has an in-situ mean direction of D=8.9°, I=46.3°, κ=9.0, α95=4.8° (n=106 samples), and a tilt-corrected one of D=280.2°, I=71.6°, κ=5.0, α95=6.8°. The fold test is negative at 95% confidence limit (in-situ ξ_2=3.41, tilt-corrected ξ_2=9.49, critical value ξ=4.20 at 95% confidence) using McFadden's (1990) method, indicating that component A is of post-folding origin.

Component B isolated from 74 samples clusters with a northwesterly shallow upward inclination and its antipode after tilt correction, and has a mean direction of D=325.7°, I=-19.9°, κ=12.1, α95=12.4°. This component is of both normal and reversed polarity although the reversed polarity is dominant (Fig. 4). The F-test (McFadden and Jones, 1981) on normal and reversed directions confirms that they share a common mean at the 95% confidence level. Application of McFadden and McElhinny's (1990) classification of the reversal test shows a class C reversal test (γ=15.9°<18.5°, critical value at 95% confidence). The fold test is positive at the 95% confidence level for component B by McFadden's (1990) new fold test (in-situ ξ_2=4.890, tilt-corrected ξ_2=2.398, critical value ξ=4.20 at 95% confidence).

Discussion

The fold test indicates that component B was acquired before folding whereas component A was obtained after folding. The corresponding in-situ VGP of the component A, at 244.9°E, 77.8°N, is not statistically different from those of Middle-Late

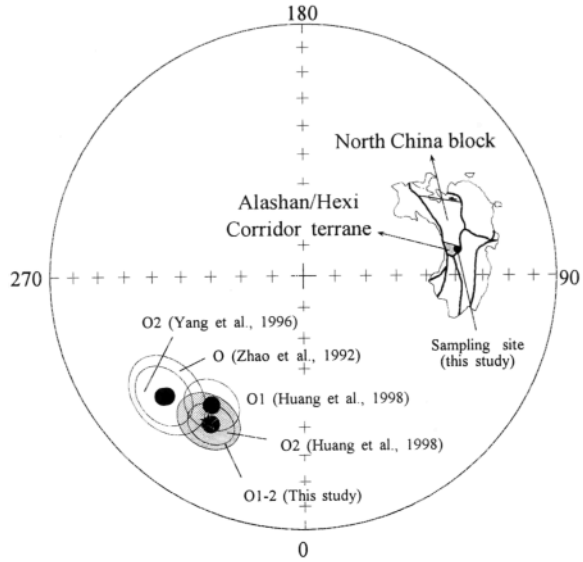


Figure 5. Equal-area projection of the Early-Middle Ordovician pole from the AHC terrane (star) and those from the sable NCB (circles). Solid/open symbols represent paleopole positions plotted onto the lower/upper hemisphere with 95% confidence limits. O represents Ordovician with 1=Early, 2=Middle.

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Jurassic or Early Cretaceous for the NCB (Yang et al., 1992) at the 95% confidence limit, indicating that component A is probably a Late Mesozoic overprint.

The pre-folding nature and the positive reversal test of component B suggests that it may be a primary magnetization acquired at or close to the time of formation of the rocks. Polarity assignment of this characteristic component is not a simple matter. Recently, Pavlov and Gallet (1998) suggested that a very low magnetic reversal frequency, with a strong reversed polarity bias is quite certain in the Early-Middle Ordovician (Arenig and Llanvirn stages). This is consistent with the Ordovician results obtained from the NCB (Yang et al., 1996; Huang et al., 1999). Huang et al. (1999) therefore assigned the northwesterly direction with upward shallow inclination as having been acquired during a normal polarity period. The Cambrian and Ordovician North poles of the NCB fall in the present Atlantic Ocean, which is contrary to the definition of Zhao et al. (1992, 1993). The Early-Middle Ordovician paleopole calculated from component B is at 326.5°E, 31.8°N with A95=9.5°. This pole is identical with those of Early-Middle Ordovician obtained from the stable NCB (Zhao et al., 1992; Yang et al., 1996; Huang et al., 1999) at the 95% confidence level (Fig. 5). This indicates that no relative movement occurred between the AHC terrane and the stable NCB since that period. The AHC terrane would be the western extension of the NCB during the Early-Middle Ordovician. This configuration is in agreement with some geological evidence that suggest the AHC terrane has been the western extension of the NCB since the Late Cambrian. Firstly, like the other parts of the NCB, the slightly metamorphosed pre-Cambrian basement and the Lower Cambrian sediments of the AHC terrane were mostly erased owing to continental uplift and denudation (Wu et al., 1993). Secondly, the most widely recognized regional unconformity in the stratigraphic sequence of the AHC terrane is observed between the Middle and Late Cambrian, which is also the case in the western Ordos Basin, suggesting that the Alashan/Hexi Corridor area might be the same tectonic unit as the Ordos Basin in that time (Zhao et al., 1993). Finally, the great stratigraphic hiatus from the Upper Ordovician through Lower Carboniferous in most areas of the NCB is also observed in the AHC terrane, where the upper Middle and Upper Ordovician series are absent (NBGMR, 1990). This indicates the AHC area may have experienced the same continental uplift and denudation as the stable NCB during the Paleozoic. The disagreement in age of the stratigraphic hiatus is probably attributed to the directional change of sea regression and transgression over the NCB (Yang et al., 1986), and the Silurian and Devonian rocks in the AHC area are mostly composed of epicontinental or continental sediments.

In conclusion, the Early-Middle Ordovician paleomagnetic result from the AHC terrane is insignificantly different from those of the stable NCB. This implies that the AHC terrane was not subject to relative movement with respect to the stable NCB since that period, and had been the western extension of the NCB during the Early-Middle Ordovician. The Silurian and Devonian paleomagnetic results from the AHC terrane may therefore be used to supply the paleomagnetic database of the NCB (Zhao et al., 1993).

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